

Homework 3: Due Monday, May 4, 2026 by 11:59pm**Please read these instructions to ensure you receive full credit on your homework.**

Submit the written portion of your homework as a *single* PDF file through Courseworks (less than 5MB). In addition to your PDF write-up, submit all code written by you in their original extensions through Courseworks (e.g., .m, .r, .py, etc.). Any coding language is acceptable, but your code should be your own. Do not submit Jupyter or other notebooks, but the original source code only. Do not wrap your files in .rar, .zip, .tar and do not submit your write-up in .doc or other file type. Your grade will be based on the contents of *one* PDF file and the original source code. Additional files will be ignored. We will not run your code, so everything you are asked to show should be put in the PDF file. Show all work for full credit.

Late submission policy: Late homeworks will have 0.1% deducted from the final grade for each minute late. *Your homework submission time will be based on the time of your **last** submission to Courseworks.* Therefore, do not re-submit after midnight on the due date unless you are confident the new submission is significantly better to overcompensate for the points lost. You can resubmit as much as you like, but each time you resubmit be sure to upload **all** files you want graded! Submission time is non-negotiable and will be based on the time you submitted your last file to Courseworks. The number of points deducted will be rounded to the nearest integer.

Problem Set-up

You are given observations $X = \{x_1, \dots, x_n\}$ where each $x_i \in \{0, 1, 2, \dots, 20\}$. You model this as being generated from a binomial mixture model of the following form:

$$x_i | c_i \sim \text{Binomial}(20, \theta_{c_i}), \quad c_i \stackrel{iid}{\sim} \text{Discrete}(\pi).$$

In this homework, you will implement two algorithms for learning this mixture model, one using maximum likelihood EM and one using variational inference. Use the data provided.

Problem 1. (35 points)

In this problem you will derive and implement an EM algorithm for learning maximum likelihood values of π and each $\theta_k \in [0, 1]$ for $k = 1, \dots, K$. Use c as the marginalized variables. In your algorithm, initialize π as a uniform vector and each θ_k so they are evenly spaced between $[0, 1]$.

- a) Show your derivation of the E and M steps of the algorithm and summarize the algorithm in pseudo-code. Since you will need to use the log marginal likelihood to assess convergence in part (b), write that out here as well.
- b) Implement the EM algorithm and run it for 50 iterations for $K = 3, 9, 15$. In one figure, plot the log marginal likelihood for iterations 2 to 50 for each K .

- c) For the final iteration of each model show the following: Construct a $K \times 21$ matrix M_K where $M_K(k, i) = q(c = k)$ learned when $x = i - 1$. You will use $p(c = k|x = i - 1, \pi, \theta)$ to find this. Show this as three images with accompanying colorbar legend for $K = 3, 9, 15$. Hint: Since the data set contains instances of all 21 possible values of x , you have already calculated the 21 required q distributions during inference.

Problem 2. (40 points)

In this problem you will implement a variational inference algorithm for approximating the posterior distribution of the binomial mixture model. As additional priors, let

$$\pi \sim \text{Dirichlet}(\alpha), \quad \theta_k \stackrel{iid}{\sim} \text{Beta}(a, b),$$

Set $\alpha = 1/10$, $a = 0.5$ and $b = 0.5$. Approximate the full posterior distribution of π , θ and $\mathbf{c}$ with the factorized distribution $q(\pi, \theta, \mathbf{c}) = q(\pi) [\prod_{k=1}^K q(\theta_k)] [\prod_{i=1}^n q(c_i)]$ using variational inference.

- a) Derive the optimal $q(\pi)$, $q(\theta_k)$ and $q(c_i)$. Also, derive the variational objective function, which will be used in part (b). Summarize the VI algorithm in pseudo-code.
- b) Implement this VI algorithm. Initialize your algorithm so that $q(\pi)$ encourages a uniform distribution and the variational parameters of all $q(\theta)$ are each initialized randomly between $[10, 100]$. Update $q(c)$ first. For $K = 3, 9, 15$ run your algorithm 10 times for 500 iterations each time and save the run with the highest variational objective function. You will therefore store one best run for each value of K . In one figure, plot the variational objective function for iterations 2 to 500 using the best run for each value of K .
- c) This part is similar to Problem 2c. For the final iteration of each saved model show the following: Construct a $K \times 21$ matrix M_K where $M_K(k, i) = q(c = k)$ learned when $x = i - 1$. Note that this q is learned differently from the EM algorithm. Show this as three images with accompanying colorbar legend for $K = 3, 9, 15$ using the best runs you saved from part (b). Hint: Since the data set contains instances of all 21 possible values of x , you have already calculated the 21 required q distributions during inference.